

ROHIT RAVINDRA POKLE

Firmware Developer | Embedded Systems & IoT Specialist

Driving scalable, high-performance firmware solutions with expertise in BLE, Matter, ESP32, and end-to-end embedded systems integration

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Profile Summary

- Firmware Developer with **2+ years of experience** in designing, developing, and optimizing embedded software for **IoT and consumer electronic devices**, delivering robust, scalable & high-performance solutions across multiple product lines.
- **Insightful** in implementing and optimizing **BLE, BLE Mesh, Matter, UDP, and MQTT protocols**, enabling **secure, low-latency, and interoperable device communication** across mobile, cloud, and embedded platforms.
- **Deep expertise in C, Embedded C, Python, and ESP-IDF**, with hands-on experience on **ESP32C3 and ESP32S3 microcontrollers**, contributing to faster product development cycles & improved firmware efficiency.
- **Resourceful in full lifecycle firmware development**, including driver integration, hardware-software interfacing, and OTA firmware updates, achieving **reliable device performance and seamless remote upgrades**.
- **Proficient in designing automation scripts, log analyzers, and debugging tools**, reducing validation time by **~20–25%** and enhancing QA testing accuracy.
- **Collaborative in partnering with cross-functional teams**—QA, cloud, and mobile developers—to ensure **end-to-end system integration**, on-time delivery, and high-quality product releases.

Core Competencies

- Embedded Systems Design & Development
- IoT Product Firmware Engineering
- Microcontroller Integration & Hardware Interfacing
- Protocol Implementation Communication (BLE, Matter, UDP, MQTT)
- Performance Optimization & Power Management
- System Integration & End-to-End Solution Delivery
- Over-the-Air (OTA) Firmware Management
- Driver Development & Device Control
- Firmware Debugging & Validation
- Automation & Testing Frameworks
- Cross-Functional Team Collaboration
- Product Lifecycle Management

Technical Skills

Programming Languages	C, Embedded C, Python (basic), C++ (basic)
Embedded Development Tools	ESP-IDF, VS Code, Arduino IDE
Protocols & Communication	UART, SPI, I2C, BLE, BLE Mesh, Matter, TCP/IP, UDP, MQTT
Debugging & Testing Tools	Wireshark, Serial Monitor (Tera Term, MobaXterm), Charles Proxy, Custom Python log visualizers
Version Control	Git
Operating Systems / Platforms	Linux (Ubuntu), macOS, Oracle VM
AI / Data Visualization	ChatGPT, TensorFlow Lite, Matplotlib
Hardware Tools	Oscilloscope, Multimeter, Soldering Station, Logic Analyzer

Domain Exposure

IoT | Consumer Electronics | Smart Home Devices | Electric Vehicles | Embedded Systems Development | Firmware Engineering

Work Experience

Aug'23 – Present | Ayla Networks India Pvt. Ltd., Bengaluru, India | Firmware Engineer

- **Embedded Firmware Development & Ownership:** Engineered and maintained firmware for **10+ smart home devices**—LED strips, humidifiers, and smart plugs—using **ESP-IDF** and **C**, ensuring robust performance and high reliability.
- **Protocol Integration & Interoperability:** Integrated **BLE Mesh** and **Matter protocols** across multiple device platforms, enabling seamless connectivity, advanced control, and cross-device interoperability.
- **Secure OTA Firmware Deployment:** Designed and implemented **OTA firmware update solutions** for ESP32S3 devices, supporting secure, efficient remote upgrades and minimizing operational downtime.
- **Automation & Validation:** Developed **Python-based automation scripts** and log analyzers to visualize real-time data, accelerating validation cycles, debugging, and QA support.
- **Cross-Functional Collaboration & Knowledge Transfer:** Partnered with **cloud, mobile app teams** for end-to-end integration, created debugging tools and training modules for QA, and ensured smooth workflow across development and testing.

Sep'22 – Nov'22 | Palis Eco Vehicle, Bengaluru, India | Embedded Trainee Engineer

- **Microcontroller & Motor Driver Expertise:** Gained hands-on exposure to **microcontroller interfacing**, motor drivers, and **power optimization strategies** for electric vehicle systems, enhancing system efficiency and reliability.
- **Embedded Code Development & Debugging:** Contributed to **embedded software debugging** for in-house vehicle control units, ensuring accurate logic execution and robust performance under varied operating conditions.
- **Hardware Validation & Testing:** Supported **hardware validation** efforts for vehicle control units, identifying and resolving functional discrepancies to ensure seamless integration with EV systems.

Projects Undertaken

- **Smart Lighting Control System | Role:** Firmware Developer | **Technologies:** ESP32H2, BLE Mesh, OTA Updates
Overview: Built embedded firmware for BLE Mesh-based light fixtures communicating with a central Transformer unit. Implemented OTA functionality and supported real-time command/status exchange via BLE Mesh.
- **CTC Humidifier | Role:** Firmware Developer | **Technologies:** ESP32, UDP, Cloud Integration
Details: Designed UDP-based firmware communication module for efficient, low-latency control from the cloud. Improved protocol error handling and ensured stability across Android, iOS, and Linux platforms.
- **Matter Protocol Testing | Role:** Firmware Developer | **Technologies:** Matter Protocol, TH Protocol Stack
Details: Conducted certification testing for Matter protocol, ensuring secure device communication and compliance with protocol standards.
- **Xtreme Light System | Role:** Firmware Developer | **Technologies:** Python, Serial Communication
Details: Led design and validation of Python-based serial log analyzer integrated with embedded logs to support visual QA tracking and improve debugging efficiency.

Education

- **2021 – 2023 |** The Oxford College of Science, Bengaluru, India | Master's in Electronics | CGPA: 7.8
- **2018 – 2021 |** MES M.M. Arts and Science College, Sirsi, India | Bachelor's in Electronics | 76.6%
- **2016 – 2018 |** Y.T.S.S, Yellapur, India | Pre-University (PCMB) | 80.76%

Personal Details

Date of Birth: 4th May 2005

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